

# AI research card

Keep this beside you when you use AI on a research question. It keeps the judgment with you.

## YOUR QUESTION

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## YOUR SENSE OF RESEARCH AND ITS ROUTE

Circle the phases your question will pass through, then note what skipping the others risks.



## THREE HABITS THAT PROTECT YOUR JUDGMENT

### Generate first

**Do:** Write a three-sentence answer or a prediction first. Then ask the AI what you missed.

**Ask the AI:** “Here is my draft answer. What did I miss, and which of my claims is weakest?”

**Why:** In a field trial, unrestricted AI raised practice performance and lowered unassisted performance; guardrails removed the harm. (Bastani et al. 2025)

### Read the source yourself

**Do:** Before you accept a synthesis, open the top source and read the parts it is cited for.

**Ask the AI:** “For each claim, quote the sentence in the source that supports it and say where it appears.”

**Why:** Most tools read only titles and abstracts. In 6,063 claims from abstracts on AI and learning, 82.1% of harm claims reported no statistical test, against 70.8% of benefit claims. (V. N. Patel 2026)

### Check every citation

**Do:** Look up every citation, and open any you plan to repeat.

**Ask the AI:** “List each source with its DOI or link, so I can check it.”

**Why:** Published tests found 55% of GPT-3.5 citations and 18% of GPT-4 citations fabricated. (Walters & Wilder 2023)

## THE EIGHT HABITS: TICK THE ONES YOU USED LAST TIME

- ☐ **Frame the question first.** Write the question and what would count as an answer before you open the tool.
- ☐ **Read the source yourself.** Before you accept a synthesis, open the top source and read the parts it is cited for.
- ☐ **Use more than one tool.** Run the same question in at least two tools, and compare what each one found.
- ☐ **Set a budget and a stop rule.** Before you start, set the time the question is worth and a rule for stopping. Extend it only by a deliberate decision.
- ☐ **Generate first.** Write a three-sentence answer or a prediction first. Then ask the AI what you missed.
- ☐ **Check every citation.** Look up every citation, and open any you plan to repeat.
- ☐ **Look for evidence against.** Keep an evidence-against column, and ask for the strongest case against your view.
- ☐ **Plan when to look again.** Note when you will revisit the answer, and what new evidence would change it.

## TAKE-HOME MESSAGES

- Say which “research” you mean. The sense sets the route, and the route shows which phases you skip and what that risks.
- Generate first, then check: write your frame, prediction or draft before you open the tool.
- Different tools find different papers. For the same question, the five tools’ top 20 results overlapped by about 2% on average. Search more than one, and read the sources yourself.
- Check every citation. Published tests found 18% to 55% of AI citations fabricated.
- A loop is only virtuous if something in it can say no: data, a dissenter, a prediction written beforehand, or a source you opened yourself.
- Set a time budget and a stop rule before you start.

**Sources.** Bastani, H. et al. Generative AI without guardrails can harm learning: evidence from high school mathematics. *Proc. Natl Acad. Sci. USA* 122, e2422633122 (2025), doi:10.1073/pnas.2422633122. · Patel, V. N. Under-reported harm: a claim-level audit of abstract reporting on AI assistance and human learning. In *Human-AI Coevolution: Measuring Human-Agent Teams in the Agentic Era* (HAIC workshop, NeurIPS 2026; non-archival). <https://github.com/vpatel43/Effect-of-AI-Assistance-on-Learning-Claim-level-evidence-audit> (2026). · Walters, W. H. & Wilder, E. I. Fabrication and errors in the bibliographic citations generated by ChatGPT. *Sci. Rep.* 13, 14045 (2023), doi:10.1038/s41598-023-41032-5.

Make your own card: answer the three questions in “What do you mean, research?”, then choose “Make my card”.  
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Interactive Case Studies